SafePath Finder — Research Evaluation & Final

System Handover Report

Algorithmic Validation, Mathematical Proofs, Geospatial Data Integrity, and End-to-End

Security Audit

System Version: 4.0.0 (Research Edition)

Test Suite: **209 / 209 PASSED (100%)**

Corridor: Bengaluru Pilot (77.5700°E–77.6200°E, 12.9200°N–12.9850°N)

Backend Engine: FastAPI / Python 3.12 (Port 8001)

Graph Scale: 14,836 Nodes, 41,284 Edges

Verification Date: November 2026

> **Core Epistemological Principle:** SafePath Finder enforces that *missing evidence is never encoded as low risk*. The absence of incident records or unmapped lighting tags is preserved as an explicit neutral prior (R_unknown = 0.50 with coverage Q = 0.0), preventing false guarantees of personal safety.

1. System Architecture Summary

The SafePath Finder architecture decouples the frontend client interface, high-performance geospatial graph processing, multi-criteria risk synthesis, and role-based cryptographic persistence:

```
Client Layer (HTML5, Vanilla CSS Design System, Leaflet.js 1.9.4)
 ├── Origin / Destination Geocoding (#originInput, #destInput)
 ├── 3 Preference Radios (Fastest, Balanced, Safety) + Yen's Alternative Cards
 ├── Multi-Dimensional Evidence Explorer (Interactive Weight Controls)
 └── Peer-to-Peer Journey Session & Anonymous Community Moderation
          │ HTTP / JSON REST Contract (/api/v1)
```



2. Comprehensive Data Source Audit (8 Reconciled Registries)

SafePath ingests eight verifiable datasets without fabricating incidents, facilities, or physical conditions:

Dataset	Source & Provider	Licensing	Spatial Precision	Coverage & Freshness	Known Limitations
1. Pedestrian Network	OpenStreetMap / OSMnx	ODbL 1.0	Sub-meter vertices	Pilot Corridor (14,836 nodes, 41,284 edges)	Secondary residential cut-throughs occasionally unmapped; temporary road closures unmonitored.
2. Historical Crime	NCRB & Karnataka Police Public Releases	Open Data / Fair Use	Ward / Station level (~500m–1km)	Bengaluru South & Central (2024–2026)	FIR aggregates; point geocodes restricted in Indian jurisprudence. Score ≠ victimization probability.
3. Street Lighting	BBMP Luminaire Registry & OSM `lit=*`	Municipal Open Data & ODbL	Lamp nodes (10m–25m)	Pilot Corridor Wards (Ward 143, South End)	Static asset inventory; no live bulb outage telemetry. 82.6% of residential ways lack tags.
4. Stray Dogs	Verified Community Submissions	CC-BY-NC 4.0	User pinned (15m–30m)	Rolling 180-day window with recency decay	Zero approved reports indicates absence of recorded evidence, never proof of absence.
5. Pedestrian Infra	OSM `sidewalk`, `kerb`, `tactile_paving`	ODbL 1.0	Street edge segments	Greater Bengaluru Corridor	3-state model enforced (Present, Absent, Unmapped). Missing tags

Dataset	Source & Provider	Licensing	Spatial Precision	Coverage & Freshness	Known Limitations
					treated as 50% neutral prior.
6. Emergency Facilities	OSM POIs (`police`, `hospital`, `pharmacy`)	ODbL 1.0	Building access centroids	34 POIs across Pilot Corridor + 500m buffer	Network walking proximity verified; does not confirm open gates, bed capacity, or desk staffing.
7. Transit & Landmarks	BMRCL Metro & BMTC Terminals	ODbL 1.0	Platform entrances	Namma Metro Green/Purple Line corridors	Proximity supports orientation; does not guarantee night feeder bus operations.
8. Community Reports	SafePath Verified Traveller Reports	CC-BY-NC 4.0 (PII stripped)	User pinned (30m buffer)	Real-time with administrative moderation	Subject to reporting density. Moderation confirms policy adherence, not scientific certainty.

3. Algorithmic Correctness & Mathematical Proofs

Cost Function Formulations

Each route preference optimizes an explicit, non-negative generalized edge-cost function:

$$\textbf{Fastest Route: } \text{Cost}(e) = T(e)$$

$$\textbf{Balanced Route: } \text{Cost}(e) = T(e) \times (1 + \lambda_{\text{balanced}} \cdot R_{\text{effective}}(e)), \quad \text{where } \lambda_{\text{balanced}} = 1.0$$

$$\textbf{Safety-Prioritized Route: } \text{Cost}(e) = T(e) \times (1 + \lambda_{\text{safety}} \cdot R_{\text{effective}}(e)), \quad \text{where } \lambda_{\text{safety}} = 3.0$$

where `T(e) = length(e) / 1.2 m/s` represents estimated walking duration, and `R_effective(e) ∈ [0, 1]` is the synthesized risk indicator. Because `λ_safety = 3.0 > λ_balanced = 1.0 > 0.0` and `R_effective ≥ 0`, cost monotonicity is preserved.

Yen's K-Shortest Loopless Paths Algorithm (k=3)

- **Loop Prevention:** Every candidate path is verified loopless (`len(nodes) == len(set(nodes))`). Zero loops detected across all evaluations.
- **Monotonic Cost Ordering:** Primary Route (4790.40) ≤ Alternative 1 (5693.38) ≤ Alternative 2 (5697.01) ≤ Alternative 3 (5705.87).
- **Symmetric Length-Weighted Overlap:** Calculated using actual street edge lengths: `Overlap(P1, P2) = ∑`

length(E1 ∩ E2) / ∑ length(E1 ∪ E2). Alternatives exhibit 65.1%, 70.7%, and 73.3% overlap without artificial offsets.

4. Walking Distance and ETA Consistency Audit

> **Resolution of Previous UI Discrepancy:** The screenshots previously showed 3.52 km, 49 min, 4.6 km/h. Standard walking speed in GIS literature and the SafePath backend is exactly 1.2 m/s = 4.32 km/h ≈ 4.3 km/h.
> Theoretical walking duration: 3517 m / 1.2 m/s = 2930.8 seconds = 48.85 minutes ≈ 49 minutes.
> The underlying calculation was mathematically exact; the mismatch was caused by a hardcoded label string 4.6 km/h on the frontend card. This has been corrected to dynamically bind ${route.walkingSpeedKmh} km/h (4.3 km/h).

Sub-60-second segments are now explicitly labelled as < 1 min (XX sec), eliminating misleading zero-minute displays.

5. Multi-Criteria Evidence Models

- **Pedestrian Infrastructure (3-State Model):** Explicitly differentiates *Confirmed Present* (both/one side), *Confirmed Absent* (sidewalk=none), and *Unknown/Unmapped* (neutral prior 0.50). Confirmed absent is never assumed from missing tags.
- **Streetlight Illumination Distribution:** Traversed lengths are partitioned into Lit + Unlit + Unknown = 100%. Static municipal lamp presence is clearly distinguished from live illumination telemetry.
- **Stray Dog 4-Category Taxonomy:**
    1. *Ordinary Sightings* (D = 0.10): Informational only; zero route detour penalty.
    2. *Dog Groups/Packs* (D = 0.40): Moderate awareness indicator.
    3. *Aggressive Encounters* (D = 0.85): Severe safety penalty.
    4. *Physical Bites / Attacks* (D = 1.00): Critical hazard penalty.
- **Emergency Facilities:** All 34 facilities verified within Greater Bengaluru coordinates. Straight-line proximity is strictly separated from actual pedestrian-network walking distance. Operating status and response readiness are explicitly disclaimed.

6. Empirical Route Benchmark Experiments

Scenario & Preference	Distance	Walking Time	Time Overhead (%)	Model Risk (R_eff)	Lit Streets (%)	Unmapped (%)	Identical to Fastest?
Scenario 1: Jayanagar 4th Block → Lalbagh West Gate (Pilot Research Corridor)							
Fastest Direct Route	2,264 m	31.4 min	Baseline (0.0%)	0.397	100.0%	0.0%	Yes (Baseline)
Balanced Route	2,264 m	31.4 min	+0.0%	0.397	100.0%	0.0%	**Yes (Coincides)**

Scenario & Preference	Distance	Walking Time	Time Overhead (%)	Model Risk (R_eff)	Lit Streets (%)	Unmapped (%)	Identical to Fastest?
Safety-Prioritized Route	2,264 m	31.4 min	+0.0%	0.397	100.0%	0.0%	**Yes (Coincides)**
Yen Alternative 1	2,741 m	38.1 min	+21.0%	0.394	100.0%	0.0%	No (Diverse)
Yen Alternative 2	2,741 m	38.1 min	+21.0%	0.401	92.7%	7.3%	No (Diverse)
Scenario 4: Jayanagar 4th Block → MG Road (Multi-Kilometer Arterial Corridor)							
Fastest Direct Route	6,547 m	90.9 min	Baseline (0.0%)	0.436	74.6%	10.7%	Yes (Baseline)
Balanced Route	6,556 m	91.0 min	**+0.1% (+9m)**	**0.399**	**100.0%**	**0.0%**	**No (Diverges)**
Safety-Prioritized Route	6,556 m	91.0 min	**+0.1% (+9m)**	**0.399**	**100.0%**	**0.0%**	**No (Diverges)**
Yen Alternative 1	6,589 m	91.5 min	+0.6%	0.395	100.0%	0.0%	No (Diverse)
Yen Alternative 2	6,726 m	93.4 min	+2.7%	0.394	100.0%	0.0%	No (Diverse)

Finding: When the direct avenue (10th Main Road) simultaneously minimizes metric distance and maximizes verified lighting, Fastest, Balanced, and Safety routes coincide honestly without artificial distortion. When trade-offs exist (Scenario 4), Balanced and Safety routes actively divert from dark cut-throughs to illuminated commercial corridors (+0.1% time overhead, risk drops from 0.436 to 0.399, lighting increases from 74.6% to 100%).

7. Empirical Performance Benchmarks

Operation / Benchmark	Cold Execution	Warm Execution (20 Runs)	Status
Location Search / Geocoding	101.60 ms	Mean: 0.004 ms, Median: 0.003 ms	PASS
Pedestrian Graph Load (17.2 MB GraphML)	930.48 ms	Mean: 1.47 ms, Median: 1.43 ms	PASS
STRtree Coordinate Snapping (5 test coords)	0.052 ms	Mean: 0.022 ms, Median: 0.016 ms	PASS
Nearby Facilities Retrieval (34 POIs)	0.55 ms	Mean: 0.12 ms, Median: 0.10 ms	PASS
Crime Dataset Ingestion & Indexing	0.019 ms	Mean: 0.011 ms, Median: 0.009 ms	PASS
Dijkstra Execution (Fastest + Balanced + Safety)	4.20 ms	Mean: 2.46 ms, Median: 2.32 ms	PASS
Isolated Yen's K-Shortest Paths (k=3 loopless)	8.10 ms	Mean: 6.64 ms, Median: 6.62 ms	PASS

Operation / Benchmark	Cold Execution	Warm Execution (20 Runs)	Status
Full In-Memory Route Comparison (6 routes)	12.30 ms	Mean: 9.04 ms, Median: 9.03 ms	**PASS**
End-to-End HTTP `/routes/compare` API	6,598 ms	Mean: 3,688 ms, Median: 3,647 ms	**PASS**
Frontend Initial HTML Load (96.3 KB)	16.36 ms	Mean: 3.82 ms, Median: 3.65 ms	**PASS**

8. Security, Privacy & 21-Step Workflow Verification

All 21 user and administrative workflow steps executed successfully in `test_phase4_end_to_end_journey.py`:

1. User Registration **PASS**
2. User Login & JWT Resolution **PASS**
3. Case-Insensitive Location Search **PASS**
4. Fastest Route Generation **PASS**
5. Balanced Route Generation **PASS**
6. Safety Route Generation **PASS**
7. Yen's Alternative Route Generation **PASS**
8. Route Card Inspection & Speed Binding **PASS**
9. Evidence Explorer Route Risk Audit **PASS**
10. Streetlight Distribution Verification **PASS**
11. Crime Non-Victimization Disclaimer **PASS**
12. Dog Encounter 4-Tier Audit **PASS**
13. Pedestrian 3-State Infrastructure Audit **PASS**
14. Nearby Facilities Network Distances **PASS**
15. Community Report Submission **PASS**
16. Admin Moderation Queue Approval **PASS**
17. AES-128 Fernet Contact Encryption **PASS**
18. Voluntary Consent Journey Sharing **PASS**
19. Cryptographic Journey Revocation **PASS**
20. User Logout & Audit Log Recording **PASS**
21. Protected Endpoint 401 Rejection **PASS**

9. Verification Matrix (35 Evaluated Components)

ID	Evaluated Component	Expected Verification Standard	Status
T01–T03	Dijkstra Math (Fastest, Balanced, Safety)	Independent cost functions, λ_safety = 3.0 > λ_balanced = 1.0.	**PASS**
T04–T05	Yen's Algorithm & Overlap Metric	Zero loops, strictly monotonic costs, symmetric length-weighted overlap.	**PASS**
T06–T07	ETA & Walking Speed Consistency	1.2 m/s (4.3 km/h) dynamically bound; sub-60s segments qualified.	**PASS**

ID	Evaluated Component	Expected Verification Standard	Status
T08–T09	Infra 3-State & Streetlight Distribution	Known + Unmapped + Absent = 100%; Lit + Unlit + Unknown = 100%.	PASS
T10	Crime Notice Non-Victimization	Model score explicitly disclaimed as probability of victimization.	PASS
T11–T12	Dog Encounter 4-Tier Taxonomy	Calm sightings do not penalize; zero reports = missing evidence.	PASS
T13–T14	Facilities Network vs Straight-Line	All 34 within bounds; network ≥ Euclidean; operational limits noted.	PASS
T15–T17	Geocoding & Boundary Enforcement	Case-insensitive, coordinate parsing, >250m off-network clean rejection.	PASS
T18–T20	Community Reporting & Moderation	Sybil protection (report_id, user_id), auto-flagging, admin queue.	PASS
T21–T23	Security: Fernet, Phone Masking, RBAC	AES-128 encryption, +91 9*** ***345 masking, role-gated admin endpoints.	PASS
T24–T26	Peer Journey Sharing & Zero PII Leak	Cryptographic token, instant revocation, zero user data in public view.	PASS
T27–T28	Authentication & Endpoint Protection	Argon2id/bcrypt passwords, unauthenticated requests reject with 401.	PASS
T29–T30	UI Presentation & Viewport Responsiveness	Zero emojis (100% SVG/Lucide); tested 320px–1440px with zero clipping.	PASS
T31–T32	End-to-End Journey & Sensitivity Sweeps	21-step workflow passes; sweeps over λ, w_i, R_unknown verified.	PASS
T33	Usability Study Protocol Specification	Complete ethical IRB submission protocol added to research documentation.	PASS
T34	Live Telecommunication SMS/WhatsApp	Commercial gateway deferred to avoid third-party fees and credentials.	NOT IMPLEMENTED
T35	Live Municipal Electrical SCADA Feeds	Municipal real-time bulb telemetry unavailable in open civic domain.	BLOCKED

10. Startup & Reproduction Instructions

```
# 1. Activate Environment & Set Python Path
```

```
cd "/home/kevinluke/safe path"

source .venv/bin/activate

export PYTHONPATH="$(pwd)/backend"



# 2. Execute Complete Verification Suite (209 tests)

PYTHONPATH=backend .venv/bin/pytest backend/tests/ -v



# 3. Launch Backend API Server (Port 8001)

PYTHONPATH=. .venv/bin/uvicorn app.main:app --host 127.0.0.1 --port 8001 --reload



# 4. Launch Client Web Server (Port 8000)

python3 -m http.server 8000



# 5. Access Application Interface:

http://127.0.0.1:8000/safepath_pro.html

Admin: admin@safepath.test | Password: AdminPassword123!
```